Deep Learning based link prediction for Minimizing Maximum Betweenness

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NetSci 2025

Introduction

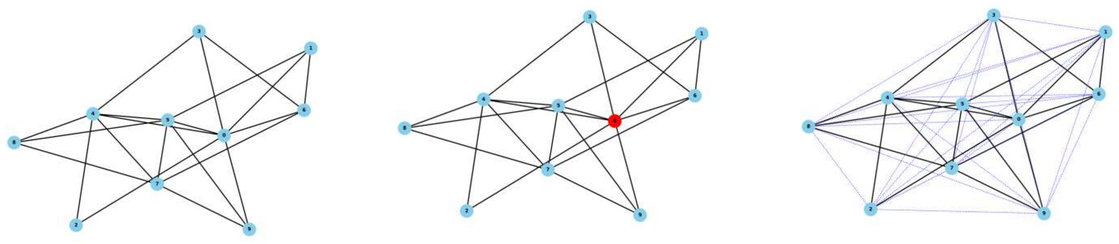
- The betweenness centrality[1,2] of a node *v* in network *G* is defined as:

$$BC_G(v) = \sum_{s,t,v \in V |\, s \neq t, t \neq v, v \neq s} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

  where $\sigma_{st}(v)$ denotes the number of shortest paths between node *s* and node *t* passing through node *v*. σ_{st} denotes the total number of shortest paths between node *s* and node *t* in network *G*.
- The highest value of betweenness centrality observed among all the nodes in a network is termed as *maximum betweenness* of the network.
- Minimizing the maximum betweenness of the network by adding missing links helps in developing better transportation systems and assists in making congestion management decisions.

Problem Definition

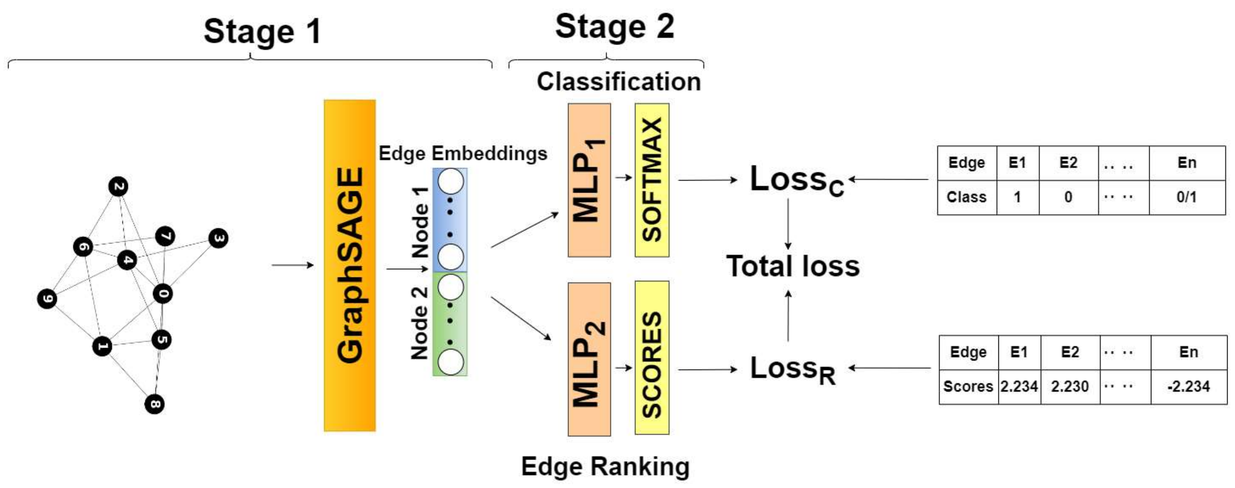
- Let *G = (V, E)* be a network, where *V* denotes the set of nodes, and *E* denotes the set of links.
- We aim to find that link $e_{min} = arg_{e \in \bar{E}} \min \left[max_{v \in V} \left[BC_{G_e}(v) \right] \right]$.



- **Goal:** Identify a missing link that, when added, minimizes the maximum betweenness centrality of the network.
- **Challenge:** High computational cost due to repeated computation of betweenness scores.
- **Proposed Solution:** Use of Graph Neural Network(GNN)-based approach to predict missing link efficiently.

Methodology

- **Two-Stage Framework**
  - **Stage1:** Use of GNN(GraphSAGE[3]) for node embedding generation via:
    - **Aggregation:** Combine neighbor features (mean aggregation used).
    - **Updation:** Update node embeddings with neighborhood and own features.
  - **Stage2:** Multi-Layer Perceptrons (MLPs) for:
    - **Classification:** Binary classification to find effective edges.
    - **Ranking:** Score and rank edges based on their potential to reduce maximum betweenness.
- **Loss Functions:**
  - **Classification Loss:** Binary cross-entropy.
  - **Ranking Loss:** Triplet margin ranking loss at place of pair-wise loss[4].



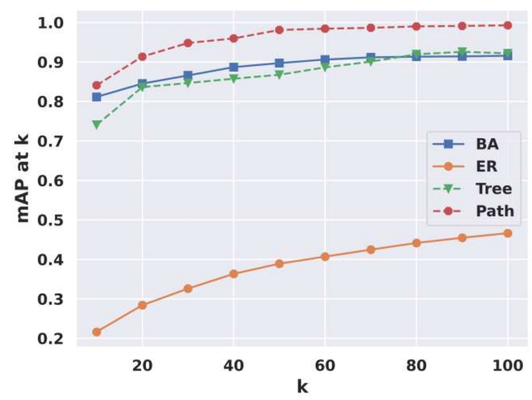
Dataset

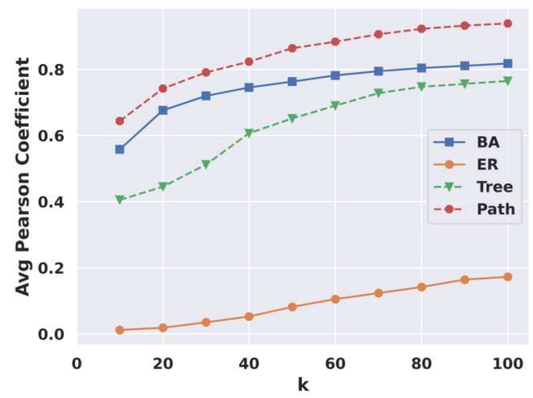
For each network, we systematically introduce missing edges into the network, one at a time, and calculate the difference between the maximum betweenness centrality of the modified network and the original network.

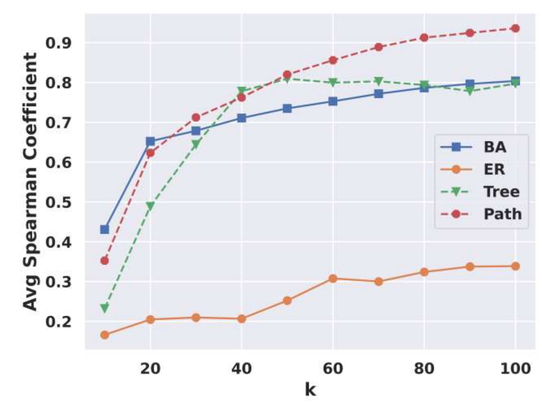
Table 1: Considered Networks

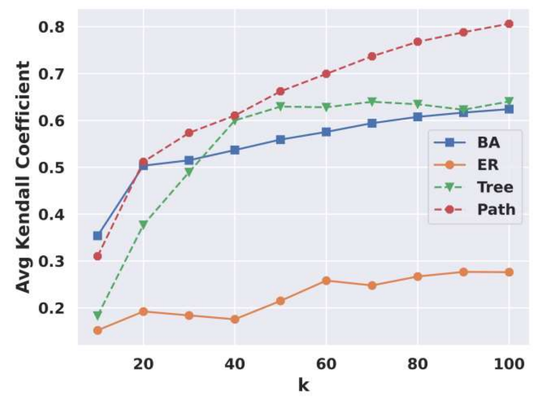
BA		ER		Tree		Path	
Parameter	**Value**	**Parameter**	**Value**	**Parameter**	**Value**	**Parameter**	**Value**
# nodes	{165, 180, 190, 200}	# nodes	{165, 180, 190, 200}	# nodes	{180, 190, 200, 210, 220}	# nodes	[160, 250]
m-value	{3, 4, 5, 6 }	Probability	[0.03, 0.0648]	k factor	{3, 4, 5, 6, 7, 8}		
seed values	[0, 1000]	seed Values	[0, 1000]				
# Graphs	100	# Graphs	100	# Graphs	30	# Graphs	10

Result

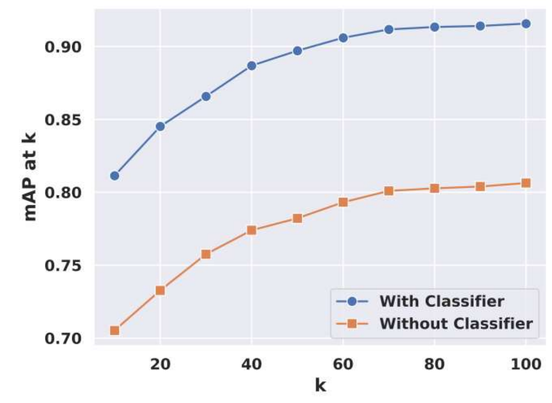


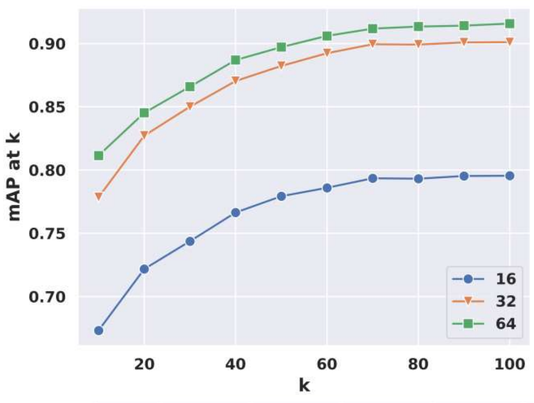


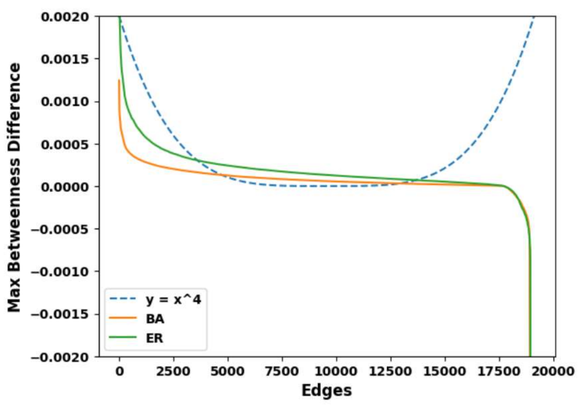


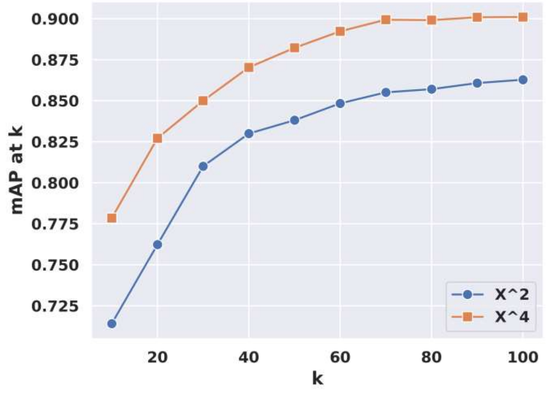


- High mAP for Barabasi-Albert networks: mAP@10=0.82, mAP@100=0.92.
- Comparable performance in tree/path networks.
- Lower performance in Erdos Renyi networks due to lack of inherent structure.









- Both MLPs are crucial for performance.
- The model performs best with a 64-dimensional edge embedding.
- Using the x^4 sampling distribution improves edge differentiation and ranking.
- Triplet margin loss outperforms pairwise ranking loss in terms of mAP.

Conclusion

- Accurate prediction for real-world-like(BA) and large-diameter networks (tree/paths).
- Demonstrates scalability and generalization for diverse network types.

References

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Aashish, Parul Diwakar, Venkata Narendra Kotyada, Rishi Ranjan Singh and Soumajit Parmanik. Deep Learning based Link Prediction for Minimizing Maximum Betweenness. Accepted and Presented at The 13th International Conference on Complex Networks and their Applications(Complex Networks 2024) Istaanbul, Turkey. (aashish@iitbhilai.ac.in)